

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 6

1 Artificial Intelligence (sbh11)

You are solving a planning problem using the GraphPlan algorithm. The problem has the following actions:

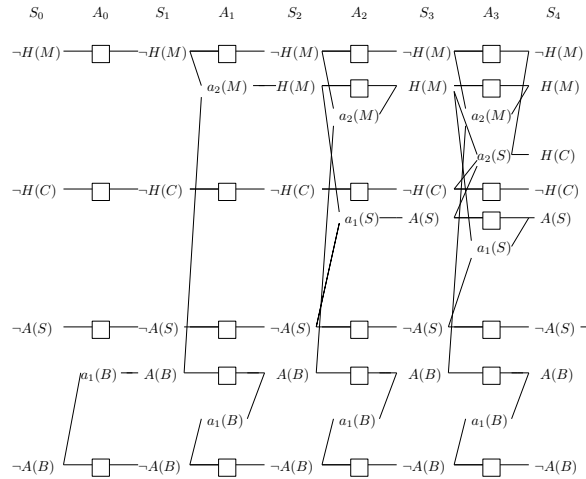
$\neg A(S), H(M)$	$\neg A(B)$	$\neg H(M), A(B)$	$\neg H(C), H(M), A(S)$
$a_1(S)$	$a_1(B)$	$a_2(M)$	$a_2(S)$
$A(S)$	$A(B)$	$H(M)$	$H(C), \neg H(M)$

The start state is $\neg H(M), \neg H(C), \neg A(S), \neg A(B)$ and the goal is $H(C)$.

Planning graphs

- (a) Draw the state levels S_i and action levels A_i of the planning graph up to and including S_2 . Do not add any mutexes at this stage. [3 marks]

Answer: The following diagram summarizes the answers for most of this question:



Note that each pair of levels A_i, S_i is a copy of the previous pair with a single added action. For this part of the question the levels S_0, A_0, S_1, A_1, S_2 are required. 1 mark for the correct organization of levels, 1 mark for the correct persistence actions, and one mark for the three actions.

Planning
graphs—plan
extraction

- (b) Explain why it is not possible at this stage to attempt to extract a plan. [1 mark]

Answer: As the goal $H(C)$ does not appear in S_2 , we know that it cannot be reached this early in the plan.

Planning
graphs—plan
extraction

- (c) What is the smallest i for which it might make sense to try to extract a plan, starting from S_i . Explain your answer. [2 marks]

Answer: As $H(C)$ first appears at S_4 we know $i = 4$. It is not in fact necessary to draw out the whole graph to establish this as it can be seen by reasoning about the actions.

— *Solution notes* —

Planning
graphs—plan
extraction

- (d) For the value of i identified in Part (c), draw levels S_{i-1} , A_{i-1} and S_i of the planning graph. [4 marks]

Answer: The levels are S_3, A_3, S_4 in the diagram above.

Planning
graphs—mutexes

- (e) On the diagram you have produced for Part (d), mark four mutexes, each of which arises for a different reason. In each case explain what kind of mutex you have included. [6 marks]

Answer: Any correctly described mutexes are acceptable. For example:

- (i) $a_2(M)$ and the persistence action for $H(M)$ compete for preconditions.
 - (ii) $a_2(S)$ and the persistence action for $\neg H(M)$ have interfering actions.
 - (iii) $a_1(S)$ and the persistence action for $\neg A(S)$ have inconsistent effects.
 - (iv) $H(M)$ and $\neg H(M)$ in S_3 are straightforward.
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GraphPlan—plan
extraction as
search

- (f) Give a general description of how the extraction of a plan from a planning graph can be addressed as a heuristic search problem. [4 marks]

Answer: The start state for the search is the set of goals for the planning problem, and a goal state for the search is a set containing the initial conditions for the planning problem. The search is then organized to work backwards through the planning graph. To move between states in the search—which are sets of unachieved conditions in the plan—a search action corresponds to a set of plan actions that achieve the conditions in the current search state, have no mutexes between them, and for which no pair of preconditions is mutex. The new search state is then the preconditions for that set of plan actions.
